

Rheological and thermodynamic behaviors of different calcium aluminosilicate melts with the same non-bridging oxygen content

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Abstract

The relationships between the chemical composition and the derivative rheological and thermodynamic values have been determined for two melt series in the anorthite–wollastonite–gehlenite (An–Wo–Geh) compatibility triangle. The melt series have 0.5 and 1 non-bridging oxygens per tetrahedrally coordinated cation (NBO/T), respectively. The influences of the ratio $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ and NBO/T on the fragility and the configurational entropy at T_g are evaluated. Linear dependencies of the viscosity, the glass transition temperature and the fragility on the ratio $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ are found for the two melt series. A crossover in the viscosity–temperature relationship is observed for both series, i.e. an inverse compositional dependence of viscosity in the high and low viscous range. The crossover presumably reflects different responses of the adjustment of melt structure to the substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} in the low versus the high viscous ranges. The crossover shifts to higher temperature with increasing NBO/T.

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1. Introduction

Understanding rheology and thermodynamics of the calcium aluminosilicate melts is important for the glass fiber industry, especially the stone wool industry, because it is beneficial to revealing the relationships between chemical composition, properties, and micro-structure of glass fibers. Studying the dependencies of fiber properties, fiber quality, fiber drawing ability on melt composition are crucial for optimizing fiber properties (e.g. bio-solubility and mechanical strength of fibers) and production conditions. The melts for making the stone fibers are multicomponent systems, each chemical component of which has a different effect on the rheological and thermodynamic properties. In this work,

the three-component aluminosilicate systems: i.e. the $\text{CaO–Al}_2\text{O}_3\text{–SiO}_2$ (CAS) systems are investigated. The three components make up approximately 80 mol% of the stone wool fibers.

Previous investigations of the compositional dependence of melt properties in the CAS system have been focused primarily on the metaluminous join $\text{SiO}_2\text{–CaAl}_2\text{O}_4$, for which the amount of non-bridging oxygen per tetrahedrally coordinated cation (NBO/T) is equal to 0. In contrast, this study is aimed at increasing the knowledge about the ‘peralkaline’ field by studying the compositional dependence of melt properties along the lines with $\text{NBO/T} = 0.5$ and 1. This will allow us to study the compositional effect on the rheological and thermodynamic properties at a constant degree of polymerization and to explore the structural changes along and between the lines. The chemical compositions of the investigated melts are chosen within the anorthite–wollastonite–gehlenite ($\text{CaAl}_2\text{Si}_2\text{O}_8\text{–CaSiO}_3\text{–Ca}_2\text{Al}_2\text{SiO}_7$) (An–Wo–Geh) compatibility triangle (see Fig. 1). So far

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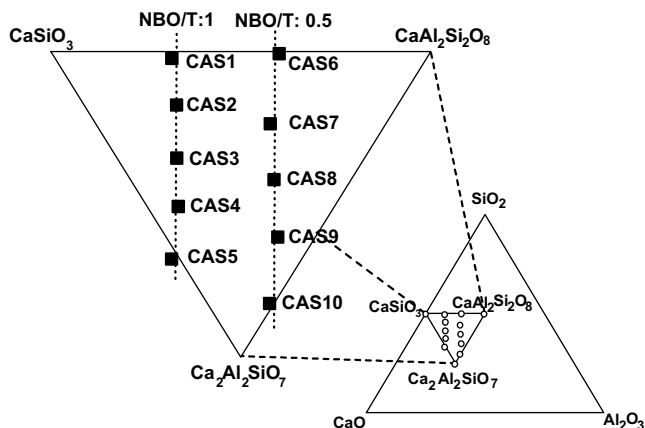


Fig. 1. The CAS system in mol%. The enlarged plot is the An–Wo–Geh compatibility triangle, which shows the composition of the 10 CAS melts.

the rheological and thermodynamic aspects of the melts in this triangle have not been systematically studied. Hence, this work will contribute to broadening and deepening the knowledge about structure and properties of high aluminosilicate melts.

The rheological and thermodynamic properties of aluminosilicate melts are determined by the arrangement of the tetrahedral structural units in the melt, which relates to the chemical bonding situation within a structural unit and between units. Aluminum differs from silicon, since tetrahedrally coordinated aluminum has to be charge-balanced by either two alkali cations or one earth alkaline cation [1]. The charge-balancing cations for the Al^{3+} tetrahedra play a large role in the melt structure. The structural role of the alkali or earth alkaline cations depends on the melt composition, i.e. whether or not Al^{3+} is present in the melt [2]. The short range ordering in the aluminosilicate network depends on the composition and the charge-balancing or network modifying cations. For aluminosilicates it is assumed that an energetically favorable case is a random occurrence of the network forming linkages Si–O–Si, Si–O–Al and Al–O–Al. The principle of Al-avoidance [3] postulates that the Al–O–Si linkage is more favorable than the combination of Si–O–Si and Al–O–Al linkages. This postulate means that the short range ordering is not random (not totally disordered). A tendency towards Al-avoidance is inferred based on ^{29}Si MAS NMR line widths in aluminosilicate glasses [4]. However the presence of a small amount of Si–O–Si in glasses of anorthite compositions observed by triple quantum MAS NMR spectroscopy [5] suggests some Al–O–Al linkages [6].

The aim of this work is to study the compositional dependencies of melt viscosity, glass transition temperature (T_g), heat capacity (C_p) as well as the derivative properties such as activation energy for viscous flow (ΔH_η), fragility [7] and configurational entropy at T_g ($S_c(T_g)$). The paper presents the results of systematic

viscometric and calorimetric experiments on the melts covered by the An–Wo–Geh compatibility triangle, and discusses the difference in the structural arrangement in the high compared to that in the low viscous range.

2. Theoretical models

Viscosity is one of the melt properties, which has been studied intensively due to its sensitivity to compositional variation [8]. At a fixed temperature it varies by orders of magnitude as a function of composition. An increase in temperature decreases the viscosity, as the structural rearrangements in the melt become easier [8].

The viscosity–temperature relationship has been described by various theories. Adam and Gibbs [9] found the correlation between the structural relaxation time of a glass forming melt and the configurational entropy (S_c). In their analysis the relaxation time increases with $1/TS_c$. The S_c decreases with decreasing temperature until it vanishes at the Kauzmann temperature. The Adam and Gibbs theory explains the temperature dependence of the relaxation in terms of the temperature dependence of the size of the cooperatively rearranging region [9].

In this work the Adam–Gibbs equation (AG equation) was fitted to experimental viscosity data to derive $S_c(T_g)$. The AG equation is expressed as

$$\log \eta = \log \eta_\infty + \frac{B}{TS_c(T)}, \quad (1)$$

where η_∞ is a pre-exponential constant and B is a constant containing a free-energy barrier which must be crossed by the rearranging group. The constant η_∞ for the glass series studied in this work was obtained by fitting the viscosity data (in the range from $\eta = 0.3$ to 10^{11} Pa s) to Eq. (1), which equals 2.88×10^{-4} Pa s [10]. The viscosity range for the validity of Eq. (1) was discussed in [11]. $S_c(T)$ can be calculated from the equation

$$S_c(T) = S_c(T_g) + \int_{T_g}^T \frac{\Delta C_p(T)}{T} dT, \quad (2)$$

where the $\Delta C_p(T)$ is the difference in heat capacity between the liquid and the glassy state. For the silicate systems, the heat capacity of the liquid state, C_{pl} , is only slightly temperature dependent and hence, can be approximately regarded as a constant. In this work, the C_p value at T_g is assigned as the heat capacity of the glassy state, C_{pg} [10]. Therefore, the $\Delta C_p(T) (= C_{pl} - C_{pg})$ can also be treated as a constant ΔC_p . Both C_{pl} and C_{pg} may be directly determined from the DSC measurement. Eq. (2) can thus be transformed into the form

$$S_c(T) = S_c(T_g) + \Delta C_p \ln \frac{T}{T_g}. \quad (3)$$

Introducing Eq. (3) into Eq. (1) leads to the equation

$$\log \eta = \log \eta_{\infty} + \frac{B}{T \left(S_c(T_g) + \Delta C_p \ln \frac{T}{T_g} \right)}; \quad (4)$$

$S_c(T_g)$ can be obtained by fitting Eq. (4) to the viscosity–temperature data. Two contributions to the $S_c(T)$ have been involved: a chemical and a topological contribution [12]. The chemical contribution is independent of temperature, whereas the relative contribution from the topological arrangement increases with temperature. Finally, at high temperatures, the chemical contribution plays a minor role. The topological arrangement is reflected by the increase of configurational entropy at T over that at T_g , and it will increase more in a relative sense for compositions having a higher ΔC_p .

Avramov [13–15] proposed a three-parameter equation that is able to predict the change in viscosity as a function of both temperature and composition. The model describes the kinetics of molecular motion in supercooled melts [13] and the dependence of the viscosity on the entropy of the melt. According to Avramov, the structural units of the system jump with frequencies depending on the activation energy, which they have to overcome. The dependence of the activation energy on temperature is assumed to obey a power decay law [13–15]. From this assumption, he proposed the equation describing the dependence of viscosity on temperature:

$$\log \eta = \log \eta_{\infty} + \varepsilon' \left(\frac{T_r}{T} \right)^{\alpha}, \quad (5)$$

where η_{∞} is a pre-exponent (in Pa s), i.e. the viscosity of a hypothetical liquid state for $T \rightarrow \infty$, ε' is the constant equal to $\log \eta_{T_r} - \log \eta_{\infty}$, α is a fragility index, and T_r is a reference temperature. In this work, T_g is used as T_r . Then, $\varepsilon' = \log \eta_{T_g} - \log \eta_{\infty}$. The viscosity of silicate melt at T_g is $\eta_{T_g} = 10^{12}$ Pa s. $\log \eta_{\infty}$ and ε' may be obtained by fitting the viscosity data to Eq. (5). For the melt series studied in this work, the optimal values of $\log \eta_{\infty}$ and ε' , which lead to best fitting, are found to be -1.7 and 13.7 , respectively, by using the relation $\log \eta_{T_g} = \log \eta_{\infty} +$

$\varepsilon' = 12$. Thus, the Avramov equation (AV equation) can be rewritten as

$$\log \eta = -1.7 + 13.7 \left(\frac{T_g}{T} \right)^{\alpha}. \quad (6)$$

The two parameters, $\log \eta_{\infty}$ and ε' , are not sensitive to the variation in chemical composition of the CAS systems. However, the sensitivity of these parameters to the variation in chemical composition has not been tested on other melt systems.

The comparisons between the AG, the AV, and the VFT (Vogel–Fulcher–Tammann) equations [16–18] were made in [10] with respect to their fitting qualities to the viscosity data for the 10 CAS melts. Then, it was found that the order of the fitting quality is $VFT < AG < AV$, the average residual errors of which were 0.116, 0.113 and 0.063, respectively [19].

3. Experimental

Ten melts covered by the $\text{CaAl}_2\text{Si}_2\text{O}_8$ – CaSiO_3 – $\text{CaAl}_2\text{SiO}_7$ (anorthite–wollastonite–gehlenite) compatibility triangle and placed along the two lines with $\text{NBO}/T = 0.5$ and 1, respectively, were synthesized (Fig. 1). The melts were synthesized from analytical chemicals SiO_2 , Al_2O_3 and CaCO_3 in a platinum crucible at 1898 K for 3 h in a MoSi_2 box furnace. Subsequently the melt was quenched on a metal plate. The chemical compositions were analyzed using an X-ray fluorescence spectrometer (XRF) (Philips 1404). The compositions are listed in Table 1.

The low viscosities ($<10^4$ Pa s) were measured by means of a concentric cylinder viscometer in the temperature range from 1316 to 1849 K at ambient pressure. The viscometer consists of four parts: a furnace, a viscometer head, a spindle and a sample crucible. The furnace was a MoSi_2 box furnace (Deltech Inc. Model DT-31-RS Mode EE) and the viscometer head was a Brookfield model RTVD with a full range torque of 7.2×10^{-2} N m, which both were controlled by a

Table 1
The composition of the 10 CAS samples

	SiO_2 (mol%)	Al_2O_3 (mol%)	CaO (mol%)	Si^{4+}	Al^{3+}	Ca^{2+}
CAS1	49.4	8.3	42.3	0.46	0.15	0.39
CAS2	45.6	10.5	43.9	0.41	0.19	0.40
CAS3	41.3	12.7	46.1	0.37	0.22	0.41
CAS4	37.3	14.7	47.9	0.33	0.26	0.42
CAS5	33.0	16.4	50.5	0.28	0.28	0.43
CAS6	49.8	15.1	35.1	0.43	0.26	0.30
CAS7	44.1	17.4	38.5	0.38	0.30	0.33
CAS8	39.5	19.9	40.5	0.33	0.33	0.34
CAS9	34.8	22.5	42.7	0.28	0.37	0.35
CAS10	29.4	24.7	45.9	0.24	0.40	0.37

The compositions are measured using XRF and given in mol% and the cation fraction.

computer. The spindle and the sample crucible were made of Pt₈₀Rh₂₀. The details about both method and equipment are described in [20]. The viscometer was calibrated using the National Bureau of Standards (NBS) 710 standard glass.

The high viscosities (10^7 – 10^{12} Pa s) were measured using the micropenetration method. The measurements were done in a vertical push-rod dilatometer (BÄHR DIL 802 V) [21]. The Deutsche Glasstechnische Gesellschaft (DGG) Standard Glass I was used for calibration. A good agreement was achieved between the measured viscosity and standard viscosity values, with the least square $R^2 = 0.996$ [10]. In the intermediate viscous range from 10^4 to 10^7 Pa s, it was impossible to measure the viscosities, because the strong tendency for the melts to crystallize hindered the measurements.

A differential scanning calorimeter (DSC) (Netzsch STA 449C) was used to determine T_g and C_p . The glass samples for the DSC measurements were the ones that had already been measured using the concentric cylinder viscometer. The samples were drilled out from the viscometer crucible wherein the glass melt was left after finishing the concentric cylinder viscometer measurements. The size of the samples for DSC measurements were $4 \times 4 \times 1$ mm. For each composition a baseline was measured with two empty crucibles. Then a standard sapphire sample was measured and finally the sample was measured in two runs. The first run was made in order to give the sample a defined thermal history, i.e. a given cooling rate that equals to the heating rate. The second run was made for determination of T_g and C_p . All measurements were done under argon with the heating rate 10 K/min. The maximum temperature of each measurement was approximately 100 K above T_g . T_g and C_p were determined from the second DSC upscan curves. T_g was defined as the onset temperature of the glass transition peak on the second DSC upscan curve.

4. Results

The viscosity–temperature relationships measured by both the micropenetration and the concentric cylinder viscometer methods are shown in Fig. 2 and also listed in Table 2. In Fig. 2, a crossover is seen for both melt series with NBO/T = 0.5 and 1, respectively. The crossover means that all viscosity–temperature curves for each melt series pass through the same viscosity–temperature point. It also means that in the highly viscous range the viscosity increases and in the low viscous range decreases with decreasing Si/(Si + AlCa_{1/2}) ratio. In other words, the compositional dependence of the viscosity for a constant temperature is opposite in the high compared to the low viscosity range. The crossovers are located at $T = 1280$ K (or

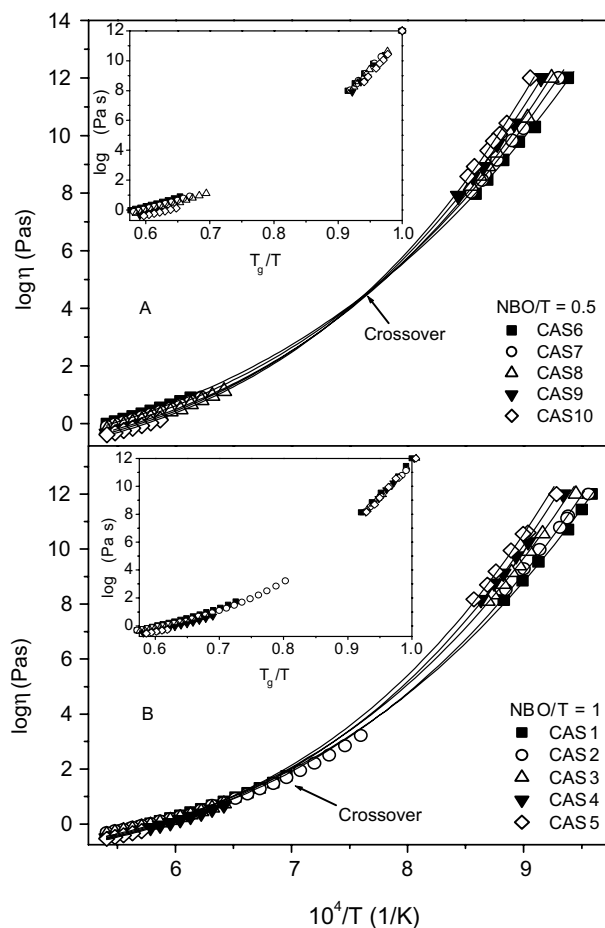


Fig. 2. Viscosity as a function of temperature. The symbols represent the measured values and the lines are the curve fits to the measured data by Eq. (6). Insets: Fragility plots. (A) NBO/T = 0.5 and (B) NBO/T = 1.

$10^4/T = 7.81$) for the line NBO/T = 0.5 and at $T = 1500$ K (or $10^4/T = 6.67$) for the line NBO/T = 1. The T_g values obtained using the DSC are listed in Table 3. Fig. 3 shows the influence of Si/(Si + AlCa_{1/2}) on the two isokom temperatures, $T_{\log \eta=12}$ and $T_{\log \eta=0}$, for the melt series NBO/T = 0.5 and 1, respectively. $T_{\log \eta=12}$ is the T_g of each melt, since generally T_g corresponds to the viscosity $\eta = 10^{12}$ Pa s. The inverse slopes shown in Fig. 3 are just a consequence of the crossover.

In classical rate theory, the activation energy may be thought of as a potential energy barrier, which is overcome by atoms when they move from one site to another in the melt [22]. Melts with high activation energy show a larger response of viscosity to a temperature change than melts with low activation energy. In this work, the activation energies for viscous flow (ΔH_η) were determined both in the glass transition range and in the high temperature range from 1750 to 1850 K (see Table 3). $\Delta H_\eta(T_g)$ and $\Delta H_\eta(\text{HT})$ refer to the activation energy around T_g and in the high temperature range, respec-

Table 2
Viscosity as a function of temperature measured on the 10 CAS melts

CAS1		CAS2		CAS3		CAS4		CAS5	
<i>T</i> (K)	log η (Pa s)	<i>T</i> (K)	log η (Pa s)	<i>T</i> (K)	log η (Pa s)	<i>T</i> (K)	log η (Pa s)	<i>T</i> (K)	log η (Pa s)
1052.7	11.44	1065.8	11.20	1091.5	10.56	1106.4	10.26	1106.5	10.61
1065.9	10.70	1066.8	11.12	1105.3	9.92	1119.1	9.75	1111.8	10.55
1095.6	9.53	1073.9	10.79	1116.1	9.40	1133.2	9.12	1124.6	9.95
1112.3	8.85	1094.3	9.98	1124.3	9.15	1143.4	8.76	1141.3	9.19
1132.6	8.14	1110.7	9.28	1133.1	8.72	1158.4	8.16	1153.1	8.70
		1130.2	8.52	1144.7	8.34			1166.7	8.17
		1316	3.22	1150.8	8.10				
		1340	2.85						
		1365	2.51						
		1389	2.21						
		1413	1.94						
1437	1.75	1437	1.69						
1461	1.51	1461	1.48						
1486	1.31	1486	1.28						
1510	1.14	1510	1.10						
1534	0.98	1534	0.93						
1558	0.82	1558	0.78	1558	0.73	1558	0.69		
1583	0.69	1583	0.68	1583	0.58	1583	0.54		
1607	0.56	1607	0.51	1607	0.45	1607	0.40		
1631	0.44	1631	0.34	1631	0.33	1631	0.28		
1655	0.32	1655	0.29	1655	0.23	1655	0.16		
1679	0.22	1679	0.19	1679	0.13	1679	0.05		
1704	0.12	1704	0.10	1704	0.05	1704	−0.05		
1728	0.02	1728	0.02	1728	−0.04	1728	−0.14		
1752	−0.07	1752	−0.06	1752	−0.11	1752	−0.22	1752	−0.26
1776	−0.15	1776	−0.13	1776	−0.18	1776	−0.30	1776	−0.34
1801	−0.23	1801	−0.20	1801	−0.24	1801	−0.37	1801	−0.41
1825	−0.31	1825	−0.25	1825	−0.31	1825	−0.44	1825	−0.48
1849	−0.39	1849	−0.31	1849	−0.38	1849	−0.50	1849	−0.53
CAS6		CAS7		CAS8		CAS9		CAS10	
1099.2	10.30	1111.2	10.26	1107.1	10.62	1121.1	10.42	1129.4	10.43
1116.2	9.79	1123.4	9.83	1140.2	9.40	1139.9	9.72	1137.2	10.05
1133.0	9.15	1140.0	9.15	1150.3	8.93	1155.8	8.91	1144.4	9.81
1151.2	8.47	1153.9	8.66	1160.4	8.60	1168.1	8.58	1150.5	9.48
1164.5	7.98	1156.9	8.45	1169.9	8.25	1185.3	7.91	1166.2	8.93
		1169.7	8.03					1173.8	8.59
				1558	1.12				
				1583	0.96				
		1607	0.91	1607	0.80				
1631	0.90	1631	0.77	1631	0.66				
1655	0.77	1655	0.64	1655	0.49				
1679	0.65	1679	0.53	1679	0.42				
1704	0.54	1704	0.42	1704	0.30			1704	0.13
1728	0.44	1728	0.32	1728	0.21			1728	0.02
1752	0.34	1752	0.23	1752	0.11	1752	−0.01	1752	−0.07
1776	0.25	1776	0.14	1776	0.03	1776	−0.10	1776	−0.16
1801	0.16	1801	0.05	1801	−0.06	1801	−0.19	1801	−0.24
1825	0.07	1825	−0.03	1825	−0.14	1825	−0.27	1825	−0.32
1849	−0.02	1849	−0.11	1849	−0.22	1849	−0.35	1849	−0.39

tively. The activation energies were obtained by finding the slope of the $\ln \eta$ vs. $1/T$ linear relationship over a narrow temperature range [10]. Fig. 4 shows linear dependences of both $\Delta H_\eta(T_g)$ and $\Delta H_\eta(HT)$ on the Si/(Si + AlCa_{1/2}) ratio.

The inset of Fig. 2 shows the viscosity–temperature relationships scaled by T_g . The fragility index α for each melt was found by fitting Eq. (6) to the log η vs. $1/T$ data

(see Table 3). Fig. 5 shows that the fragility increases with decreasing Si/(Si + AlCa_{1/2}) for both lines. In addition, it is found that α increases with increasing mol% CaO for both lines (Fig. 6). The AG equation (Eq. (4)) was used to find the configurational entropy at T_g ($S_c(T_g)$) and the results are listed in Table 3 and shown in Fig. 7, where it is seen that $S_c(T_g)$ decreases with decreasing Si/(Si + AlCa_{1/2}).

Table 3
Rheological and thermodynamic quantities of the 10 CAS melts

	T_g^a (K)	$\Delta H_\eta(T_g)^b$ (kJ/mol)	ΔH_η (HT) ^b (kJ/mol)	α^c	B^d (kJ/mol)	$S_c(T_g)^d$ (J/(mol K))
CAS1	1043	937	208	4.18	252	6.62
CAS2	1057	950	170	4.36	246	6.52
CAS3	1065	991	172	4.40	254	6.47
CAS4	1075	992	189	4.61	246	6.14
CAS5	1084	1051	175	4.62	271	6.84
CAS6	1066	881	231	3.83	302	7.77
CAS7	1075	961	219	4.05	292	7.35
CAS8	1082	939	217	4.25	284	7.09
CAS9	1093	1002	217	4.40	290	7.24
CAS10	1104	1036	213	4.56	284	7.08

^a Determined using the DSC, with the heating and cooling rates 10 K/min.

^b Estimated in the glass transition region as well as in the high temperature (HT) range 1700–1850 K.

^c Found by a least square fit of the viscosity–temperature data using the Avramov equation [10,13].

^d Obtained by fitting the Adam–Gibbs equation to the viscosity data [9,10].

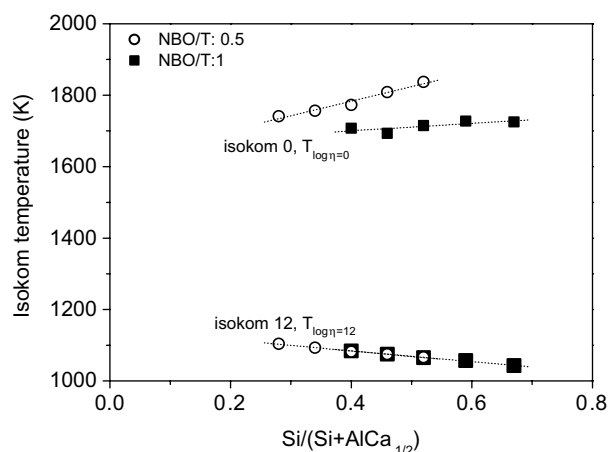


Fig. 3. The isokom temperatures, $T_{\log \eta=12}$ (i.e. T_g) and $T_{\log \eta=0}$, as functions of the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio for the melt series with $\text{NBO}/\text{T} = 0.5$ and 1. The dashed lines represent linear fits.

5. Discussion

The appearance of the crossover (see Fig. 2) and inverse compositional dependence of $\Delta H_\eta(T_g)$ and $\Delta H_\eta(\text{HT})$ (see Fig. 4) suggest that the changes in structural arrangement with chemical composition are different in the high and low viscosity ranges. The change of viscosity of a silicate melt with temperature is directly associated with that of the frequency of the breaking and bridging of $\text{T}'\text{--O}$ bonds, where T' stands for tetrahedrally coordinated cations. With increasing temperature, the breaking of $\text{T}'\text{--O}$ bonds becomes more frequent, so that the structure units become more mobile and viscosity decreases. According to [23], the changes of viscosity of a melt is also associated with the exchange between different Q^n units, where n refers to the number of the oxygens per SiO_4 or AlO_4 unit (Q), which are shared with another SiO_4 or AlO_4 unit [23]. Such exchange controls viscous flow of melts [23–25]. The higher the exchange frequency is, the lower is the vis-

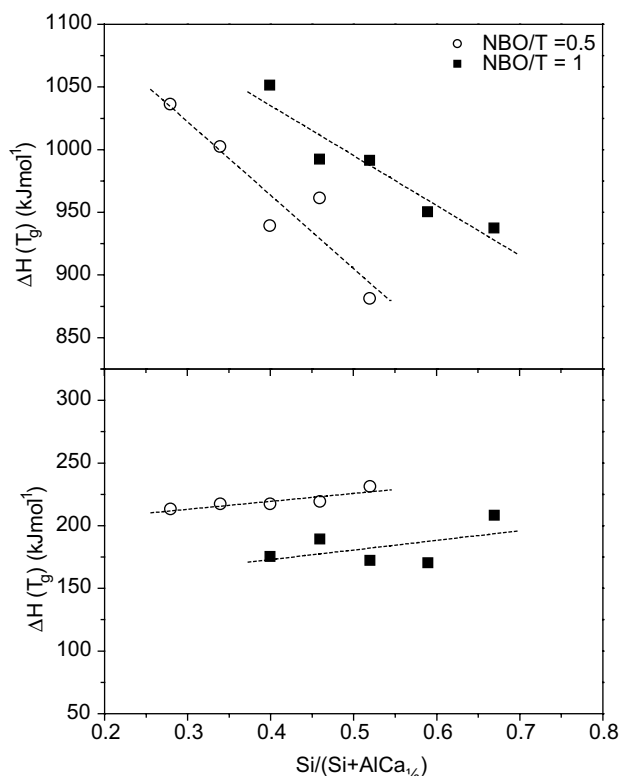


Fig. 4. The activation energy for viscous flow as a function of the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio for the melt series with $\text{NBO}/\text{T} = 0.5$ and 1, respectively. (a) The $\Delta H_\eta(T_g)$ and (b) $\Delta H_\eta(\text{HT})$, where HT refers to the temperature range 1700–1850 K. Dashed lines represent linear fits.

cosity of the melt. The exchange frequency depends on the glass composition and hence fragility of a melt. In high temperature region a fragile melt has higher exchange frequency than a strong one, whereas in low temperature region the former has lower exchange rate than the latter. Consequently, the ratio in the exchange rate between the low and the higher temperature increases with substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} . In the high viscous range, the melt structure is not per-

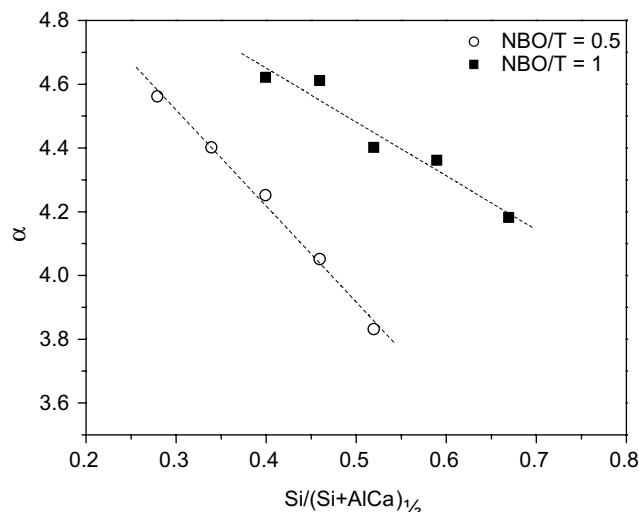


Fig. 5. The fragility index, α , as a function of the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio for the melt series with $\text{NBO}/\text{T} = 0.5$ and 1. The α values are obtained by fitting Eq. (6) to the measured viscosity–temperature relationship. Dashed lines represent linear fits.

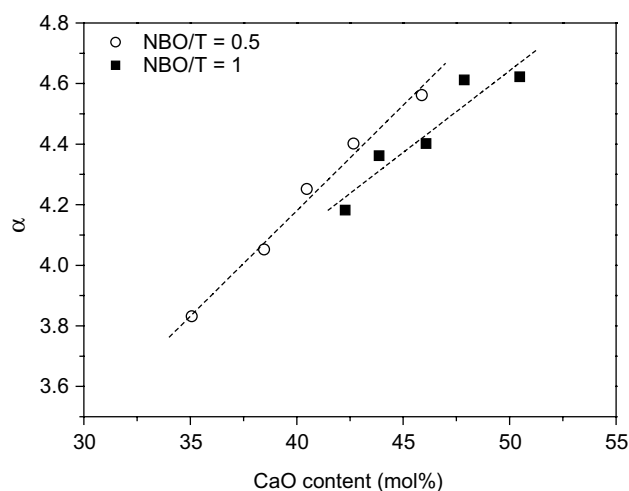


Fig. 6. The fragility index, α , as a function of the CaO content (mol%). Dashed lines represent linear fits.

turbed in the same way as in the low viscous range where the structural units of the glass melt are mobile and, hence, the diffusion is faster. If temperature is scaled by T_g , the crossover disappears as shown in the insets of Fig. 2, but the curvatures of the $\log \eta$ vs. T_g/T plots, that are quantified by the fragility index α , become different among the glasses. Thus the existence of a crossover implies the difference in the fragility between the different glasses.

The dependence of viscosity on the temperature indicates a high and a low temperature structural arrangement in the melts. The diffusivity is dramatically lower in a high viscous range compared to that in a low viscous range. In the high viscous range ($\eta = 10^8$ – 10^{12} Pa s), both the charge-balancing ions and the network-

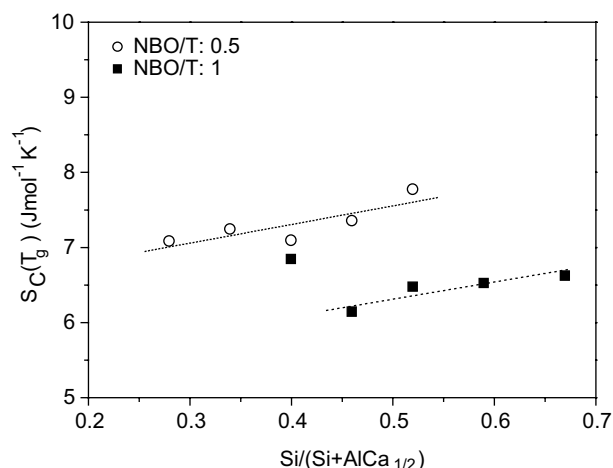


Fig. 7. The configurational entropy at T_g , $S_c(T_g)$, as a function of the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. $S_c(T_g)$ is obtained by fitting Eq. (4) to the measured viscosity–temperature relationship. The dashed lines are guides to the eye.

modifying cations (Ca^{2+}) have the tendency to attract the neighboring tetrahedra of glass former. Recently, it has been experimentally confirmed that the Ca^{2+} ions contract the channels in the glass network as reported in [26]. Therefore, the structural network becomes stronger with increasing substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} because of the role of Ca^{2+} ions. In other words, the connection of tetrahedra becomes stronger, and hence the formation of larger clusters is favored, if $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ substitutes for Si^{4+} . As a result, the viscosity increases with that substitution in the high viscous range. In this case, the topological and cooperative rearrangements of the melt structure are dominant in influencing the change in viscosity. In the high viscous range, the diffusion of the structural units (e.g. tetrahedral groups and Ca^{2+} ions) in the melt become more difficult with increasingly substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} because of the strengthening of structural network by Ca^{2+} . That is why $\Delta H_\eta(T_g)$ increases with decreasing $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ (see Fig. 4). The strengthening of the melt structure is also reflected by the increase in T_g with increasingly substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} (see Fig. 3 and Table 3). This is because a high T_g value corresponds to the high strength and/or the high degree of polymerization of the melt structure.

In the low viscous range, the strength of the chemical bonds, including both the charge-balancing ions and the network-modifying cations lying in the interstice between the Al-tetrahedrons becomes weaker and more unstable. The diffusion process of the structural units become faster and more dominant with substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} and hence leads to a decrease in viscosity for both of the NBO/T lines (see Fig. 3).

The crossover can also be explained by the recent study on difference in structural relaxation time and

calcium mobility for calcium aluminosilicate melts. In [27], it has been found that in the high temperature range the characteristic time of the calcium mobility is the same as the structural relaxation. In contrast, in the low temperature range two kinds of relaxation processes exists. Firstly, a fast process, which corresponds to calcium mobility, and secondly, a slow one, which corresponds to structural relaxation of the aluminosilicate network.

The crossover shifts to higher temperature with increasing NBO/T (see Fig. 2). In other words, the temperature range above T_g , in which a substitution of $\text{Al}^{3+} + 1/2\text{Ca}^{2+}$ for Si^{4+} increases the viscosity, becomes broader. This suggests that an increase in the concentration of Ca^{2+} results in broadening of the temperature range, in which Ca^{2+} ions strengthen the melt structure by attracting the tetrahedra and thus enhancing the grouping of these tetrahedra, i.e. the formation of the clusters. Such structure strengthening is reflected by the fact that the T_g values of the melts with NBO/T = 1 are higher than those of the melts with NBO/T = 0.5, as shown in Fig. 3 and Table 3. The shift of the crossover also implies that the topological and cooperative rearrangement of the melt structure in the high viscous range is more predominant and the formation of cluster is more favored for the aluminosilicate melt with more Ca^{2+} content than for the melt with less Ca^{2+} content, whereas the diffusion of the structural units become more hindered for the former than the latter.

The fragility index α shows a linear dependence on both the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio (see Fig. 5) and the CaO content (mol%) (see Fig. 6) for both melt series with NBO/T = 0.5 and 1, respectively. This clearly indicates the sensitivity of fragility to the content of Ca^{2+} in the manner that CaO greatly can increase the fragility of the studied glass melt system. From Fig. 5, it can also be seen that the fragility changes more dramatically for melts with NBO/T = 0.5 compared to that of melts with NBO/T = 1.

In Fig. 8, the configurational entropy $S_c(T)$ is compared between different melts for a certain value of T/T_g . At $T/T_g = 1.1$, $S_c(T)$ slightly increases with increasing $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio, whereas at $T/T_g = 2$, $S_c(T)$ decreases with increasing $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. This shows that the dependence of $S_c(T)$ at $T/T_g = 1.1$ on the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio is inverse to that for $T/T_g = 2$. In other words, the compositional dependencies of $S_c(T)$ at $T/T_g = 1.1$ and at $T/T_g = 2$ are inverse. This indicates a relative change of the ratio between the chemical and topological contributions to the configurational entropy. With increasing temperature, the relative chemical contribution to $S_c(T)$ decreases, while the topological contribution increases (see Eq. (4)). The relative increase in $S_c(T)$ with increasing temperature is largest for the melt with the lowest $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio (see Fig. 8), because it has the highest ΔC_p (see Eq. (4)) [10]. This is

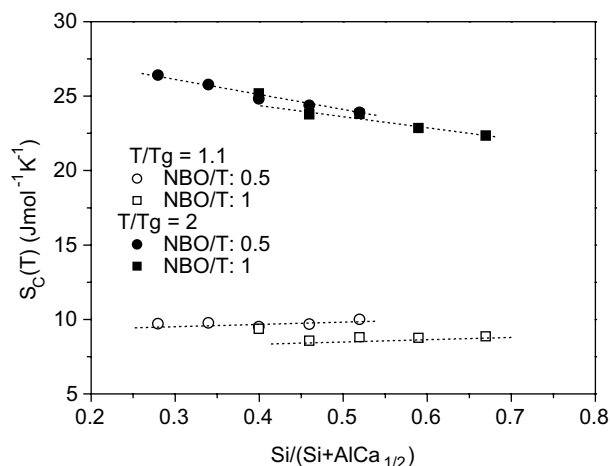


Fig. 8. Configurational entropy as a function of the ratio $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ at $T/T_g = 1.1$ and 2.0 , respectively. Dashed lines represent linear fits.

related to the fact that the perturbation of the local structural environment becomes more intense with decreasing the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. The increase in concentration of Al^{3+} leads to an increase in degree of disorder of the melt structure. In the low temperature range, the chemical contribution controls the configurational entropy, whereas in the high temperature range the topological contribution controls the configurational entropy as also discussed in the work [12,28].

Regarding each of the two compositional joins (NBO/T = 0.5 and 1) as part of a mixing series, a negative deviation from linearity in T_g values from the end members is anticipated. In an ideal mixing series, a minimum occurs when half of each end member is mixed. This is analogous to the phenomenon that is clearly seen when dealing with the mixed alkali effect [29]. A minimum in T_g vs. $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ for the lines NBO/T = 0.5 and 1 is observed [10,30]. The increasing amount of Al–O–Si bonds relative to that of Si–O–Si bonds results in the decrease in T_g when looking in the direction of decreasing $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ until a minimum in T_g is reached. The minimum is followed by an increase in T_g when additionally decreasing the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. This behavior is attributed to the increasing possibility for the charge-balancing cations and the network modifying cations to attract tetrahedrons. The formation of clusters becomes easier with decreasing the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. The consequence is a strengthening of the melt structure.

The occurrence of a minimum in T_g for the lines NBO/T = 0.5 and 1 can be understood as a consequence of the mixing effect in the manner that $S_c(T_g)$ depends non-linearly on the $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio, but with a maximum at a certain $\text{Si}/(\text{Si} + \text{AlCa}_{1/2})$ ratio. The mixing effect can be regarded as similar to that of the mixed alkali effect [29]. For both melt series with NBO/T = 0.5

and 1, the non-linearity between $S_c(T_g)$ and the Si/(Si + AlCa_{1/2}) ratio is not seen (Fig. 7), because of lack of a full mixing series. Both melts series all have a Si/(Si + AlCa_{1/2}) ratio, at which both the minimum in $S_c(T_g)$ and the minimum in T_g occur. The occurrences of the $S_c(T_g)$ maximum and the T_g minimum at a certain Si/(Si + AlCa_{1/2}) ratio are a response to the mixing effect. A similar and even more pronounced response to mixing is reported for the mixing series grossular-pyrope, for which Ca²⁺ is substituted by Mg²⁺ [31]. For a certain temperature in the high viscous range, the viscosity as a function of the Si/(Si + AlCa_{1/2}) ratio deviates from the linearity between the viscosity of the two end members in the mixing series. An identical trend is seen for the end members mixed for the line NBO/T = 0, i.e. the end member melts SiO₂ and CaAl₂O₄ [10].

For peralkaline low silica melts with the CaO/Al₂O₃ = 1.6, it is observed that T_g depends on the silica content [32]. For this specific compositional join, the NBO/T ratio is not constant. The dependence of T_g on the Si/(Si + AlCa_{1/2}) ratio is related to structural rearrangements in the melt, either a change in the arrangement of Ca²⁺ neighbors around Al³⁺ atoms or a change in the network polymerization [32]. When applying this consideration to the lines NBO/T = 0.5 and 1, it is found that the change in the arrangement of Ca²⁺ around Al³⁺ was responsible for the dependence of T_g on the Si/(Si + AlCa_{1/2}) ratio. It is the attraction of tetrahedrons by the charge-balancing cation that causes the increase in T_g with decreasing Si/(Si + AlCa_{1/2}). At the ratio CaO/Al₂O₃ = 1.6, both the Ca²⁺ arrangement around Al³⁺ and the polymerization degree are responsible for the dependence of T_g on the Si/(Si + AlCa_{1/2}) ratio.

6. Conclusions

For melts along the lines NBO/T = 0.5 and 1 and covered by the An–Wo–Geh compatibility triangle, viscosity increases with decreasing the Si/(Si + AlCa_{1/2}) ratio in the high viscous range. In contrast, in the low viscous range, the viscosity decreases with decreasing Si/(Si + AlCa_{1/2}) ratio. This inverse compositional dependence of viscosity leads to the occurrence of a crossover in the viscosity–temperature relationship for both NBO/T lines. The crossover reflects a different response of the structural arrangement to the substitution of Al³⁺ + 1/2Ca²⁺ for Si⁴⁺ in the low compared to that in the high viscous range, i.e. a high and a low temperature structural arrangement. In the low temperature range, the strengthening of the melt structure by Ca²⁺ is responsible for the increase in viscosity with the substitution of Al³⁺ + 1/2Ca²⁺ for Si⁴⁺, whereas in the high temperature range, the enhancement of diffusion of the structural units is responsible for the decrease in viscosity

with that substitution. As a consequence, the activation energy for viscous flow near $T_g(\Delta H_\eta(T_g))$ increases with decreasing Si/(Si + AlCa_{1/2}), whereas the activation energy at high temperatures ($\Delta H_\eta(\text{HT})$) slightly decreases. Both $\Delta H_\eta(T_g)$ and $\Delta H_\eta(\text{HT})$ vary linearly with decreasing Si/(Si + AlCa_{1/2}).

The fragility of both melt series increases linearly with decreasing Si/(Si + AlCa_{1/2}). At the same ratio of Si/(Si + AlCa_{1/2}), the melt series with NBO/T = 1 is more fragile than that with NBO/T = 0.5. However, the sensitivity of the fragility to the ratio Si/(Si + AlCa_{1/2}) is higher for the former than for the latter. The configurational entropy at $T_g(S_c(T_g))$ increases with decreasing the NBO/T and the Si/(Si + AlCa_{1/2}) ratios, indicating that the degree of the structural disordering increases.

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